Factory-to-Customer Shipping Route Efficiency Analysis

for Nassau Candy Distributor

📦 Supply Chain Analytics | 🗺️ Logistics Optimization | 💻 Streamlit Dashboard

Project Type	Data Analytics + Interactive Dashboard
Tools Used	Python, Pandas, NumPy, Plotly, Streamlit, Jupyter Notebook
Live Dashboard	https://candy-shipping-analysis.streamlit.app/
Focus Area	Lead Time, Route Efficiency, Delays, Bottlenecks, Ship Mode Analysis

📝 Abstract

In modern supply chain systems, shipping efficiency plays a critical role in ensuring customer satisfaction and operational performance. This project analyzes the factory-to-customer shipping routes of Nassau Candy Distributor to identify inefficiencies, delivery delays, and logistics bottlenecks across regions. Using transactional order data, key metrics such as shipping lead time, delay frequency, route volume, and route efficiency scores were computed.

The analysis was implemented using Python, Pandas, NumPy, Plotly, and Streamlit. The final outcome is an interactive web dashboard that allows users to explore route performance, geographic delay patterns, ship mode efficiency, and high-volume bottleneck states. The findings provide actionable insights to optimize logistics operations, reduce delivery delays, and improve customer satisfaction.

🔑 Keywords

> Supply Chain Analytics, Shipping Efficiency, Route Optimization, Logistics Dashboard, Streamlit, Python, Lead Time Analysis, Route Efficiency Score, Nassau Candy Distributor

📌 1. Introduction

Efficient logistics management is a cornerstone of successful distribution systems. Companies operating at a national scale must ensure timely delivery while maintaining cost efficiency. Nassau Candy Distributor ships products from multiple factories to customers across different regions, states, and shipping modes, making route optimization essential.

Without route-level visibility, logistics decisions often remain reactive. This project provides a data-driven framework to evaluate shipping efficiency, detect bottlenecks, compare shipping modes, and support better operational decision-making through an interactive Streamlit dashboard.

❗ 2. Problem Statement

The organization lacks clear visibility into the performance of factory-to-customer shipping routes. This creates challenges in identifying inefficient routes, delay-prone regions, and operational bottlenecks.

🚚 Which shipping routes are efficient and inefficient?

🗺️ Which states or regions experience frequent delays?

📦 How do different ship modes impact delivery performance?

⚠️ Which high-volume routes need immediate logistics attention?

🎯 3. Objectives

⏱️ Calculate shipping lead time for each order.
🏆 Identify top-performing and slow-performing routes.
📊 Analyze delivery delays by state, region, division, factory, and ship mode.
🚧 Detect high-volume bottleneck states and routes.
💻 Build an interactive dashboard for business users and decision-makers.

🗂️ 4. Dataset Description

The project uses shipment and order-level data for Nassau Candy Distributor. The dataset includes order dates, ship dates, product names, customer location details, ship modes, sales, cost, gross profit, region, division, and state/province information. These fields were used to calculate lead time, create factory-to-customer routes, and evaluate logistics performance.

🛠️ 5. Methodology

Stage	Process	Output
Data Cleaning	Converted date fields, removed invalid lead times, handled missing values, and standardized fields.	Clean and reliable shipment data
Feature Engineering	Calculated lead time, mapped products to factories, and created route fields.	Factory-to-state and factory-to-region routes
Aggregation	Grouped by route, ship mode, state, region, division, and factory.	Route-level and location-level performance metrics
KPI Development	Calculated average lead time, delay frequency, route volume, gross profit, and efficiency score.	Business-ready logistics KPIs
Dashboarding	Built an interactive Streamlit dashboard using Plotly visualizations.	Web-based analytics application

📊 6. Exploratory Data Analysis

✅ Most shipments are completed within a short lead-time range after data cleaning.
📍 Delivery performance varies significantly across regions and routes.
🚚 Some factory-to-state routes show consistently low lead times.
⚡ Ship mode selection has a direct impact on delivery speed and delay frequency.

💻 7. Dashboard Implementation

An interactive Streamlit dashboard was developed to convert the analysis into an easy-to-use business intelligence application. The dashboard provides filters for date range, ship mode, region, division, state/province, delay threshold, and detailed or summary view.

📈 KPI cards showing average lead time, total shipments, delayed shipments, sales, gross profit, and efficiency score.

🏆 Route performance leaderboard showing fastest and slowest routes.

🗺️ US geographic map for state-level shipping performance.

🚚 Ship mode analysis for comparing lead time, volume, and delay rate.

🔍 Route drill-down and bottleneck analysis for operational decision-making.

📋 Executive summary with automated insights and recommendations.

> 🌐 **Live Dashboard:** https://candy-shipping-analysis.streamlit.app/

📈 8. Key Performance Indicators

KPI	Purpose
⏱️ Shipping Lead Time	Measures delivery duration from order date to ship date.
📊 Average Lead Time	Compares average delivery performance across routes and ship modes.
📦 Route Volume	Counts the number of shipments handled by each route.
⚠️ Delay Frequency	Calculates the percentage of shipments exceeding the delay threshold.
🏆 Route Efficiency Score	Ranks routes using normalized lead-time performance.
💰 Gross Profit	Measures business value associated with logistics performance.

🔍 9. Results and Key Insights

🚚 **Route Performance Variation:** Some factory-to-state routes consistently deliver faster, indicating efficient logistics networks.

🗺️ **Regional Bottlenecks:** Certain regions and states show higher lead times, suggesting distance, routing, or infrastructure-related challenges.

🚧 **High-Volume, High-Delay Routes:** Routes with both high shipment volume and long lead times are critical bottlenecks that require priority action.

⚡ **Shipping Mode Impact:** Expedited shipping methods generally reduce delivery time, while standard shipping may offer cost advantages but longer lead times.

🗺️ 10. Geographic Bottleneck Analysis

The geographic analysis identifies states and regions with poor shipping performance. By combining shipment volume and average lead time, the dashboard highlights locations where logistics optimization is most needed. High-volume states with long lead times should be prioritized for route redesign, regional distribution support, or improved carrier planning.

🚚 11. Ship Mode Performance Analysis

Shipping modes were compared based on average lead time, shipment count, delay rate, cost, and sales performance. Faster shipping options improve delivery reliability but may increase cost. Standard shipping remains useful for cost-effective logistics when delivery urgency is lower. The dashboard helps compare these trade-offs interactively.

💡 12. Recommendations

🚧 Optimize high-delay routes with both high volume and long lead times.
🏭 Introduce or improve regional distribution strategies to reduce distance-related delays.
⚡ Use expedited shipping strategically for priority or delay-prone shipments.
📊 Monitor route efficiency KPIs regularly through the Streamlit dashboard.
🧭 Improve route planning using state, region, factory, and ship mode insights.

🔮 13. Future Scope

🤖 Predictive modeling for delivery time estimation.
🛣️ Distance-based route optimization.
📡 Real-time logistics monitoring dashboard.
🔗 Integration with live carrier and warehouse data.
🧠 Machine learning model for delay prediction and risk scoring.

✅ 14. Conclusion

This project provides a comprehensive, data-driven analysis of shipping route efficiency for Nassau Candy Distributor. By transforming raw shipment data into actionable insights, the organization can proactively address inefficient routes, reduce delays, and improve customer satisfaction.

The Streamlit dashboard strengthens the project by making the analysis interactive and decision-oriented. It allows users to explore route performance, geographic patterns, ship mode efficiency, and bottlenecks in real time. Overall, the project demonstrates how data analytics and dashboarding can improve supply chain decision-making and operational performance.

🔗 15. Project Links

🌐 **Live Streamlit App:** https://candy-shipping-analysis.streamlit.app/

💻 **GitHub Repository:** https://github.com/Prasannasegabandi36/Nassau-Candy-Distributor

📁 **Dashboard File:** candy_app.py